VALIDATION PLAN
OPERATIONAL QUALIFICATION
PROCESS CAPABILITY MODULE

JR Validated Environment — Process Capability Module v1.0

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Version History

Version	Date	Author	Description
1.0	2026-03-24	Joep Rous	Initial release. Covers all 3 Cap scripts in module v1.0.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-24
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the Process Capability module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each Process Capability script performs correctly for its intended use.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the Process Capability module.

2. Scope

2.1 In Scope

- All 3 R scripts in repos/cap/R/ (jrc_cap_normal, jrc_cap_nonnormal, jrc_cap_sixpack)
- Correct computation of Cp, Cpk, Pp, Ppk, and Cpm for normally distributed individual data
- Correct computation of Pp and Ppk via the percentile method for non-normal data
- Correct construction of the six-panel capability report (I-MR chart, histogram, Q-Q plot, numerical summary, verdict panel)
- Correct rejection of invalid inputs with informative error messages (missing file, n < 5, non-numeric column, LSL >= USL, both spec limits omitted)
- Correct file I/O behaviour (CSV input, PNG output to ~/Downloads/)
- Verification that all R scripts enforce execution within the jrrun validated environment by checking RENV_PATHS_ROOT at startup

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Performance qualification (PQ) — process capability studies using real production data are conducted separately under individual process validation protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2, grid) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying methods (moving range sigma estimation, percentile method) — method references are cited in Section 3
- jrc_capability.R (community script, separate OQ coverage under JR-VP-002)

3. Regulatory Basis and References

3.1 Regulations

Reference	Title
21 CFR Part 820, §820.70(i)	Quality System Regulation — Automated Data Processing
21 CFR Part 820, §820.75	Process Validation
21 CFR Part 11	Electronic Records; Electronic Signatures
ISO 13485:2016, §7.5.6	Validation of processes for production and service provision

3.2 FDA Guidance

Reference	Title	Date
FDA GPSV	General Principles of Software Validation; Final Guidance for Industry and FDA Staff	January 2002
FDA CSA	Computer Software Assurance for Production and Quality System Software; Draft Guidance	September 2022

3.3 Standards and Methods

Reference	Title
AIAG SPC Manual, 2nd Ed. (2005)	Statistical Process Control Reference Manual. Automotive Industry Action Group. Moving range sigma estimation (d2 constant); capability index formulas.
ISO 22514-2:2017	Statistical methods in process management — Capability and performance. Part 2: Process capability and performance of time-dependent process models.
ISO 22514-1:2014	Statistical methods in process management — Capability and performance. Part 1: General principles and concepts.
ISO 13485:2016	Medical devices — Quality management systems — Requirements for regulatory purposes.
Kane (1986)	Kane VE. Process capability indices. Journal of Quality Technology, 18(1), 41–52. Original publication of Cpk index.
Taguchi et al. (1989)	Taguchi G, Elsayed EA, Hsiang TC. Quality Engineering in Production Systems. McGraw-Hill. Cpm index formulation.
Shapiro & Wilk (1965)	Shapiro SS, Wilk MB. An analysis of variance test for normality (complete samples). Biometrika, 52(3-4), 591–611.

3.4 Internal Documents

Document Number	Title
JR-VP-001	Validation Plan — Installation Qualification, JR Validated Environment v1.0.0
JR-IQ-001	IQ Execution Evidence (docs/IQ_validation_20260311_205146.txt)
JR-VP-002	Validation Plan — OQ, Community Script Suite v1.1.0

4. Definitions and Abbreviations

Term	Definition
Cp	Process capability ratio — spread-only index. Cp = (USL − LSL) / (6σ_w). Requires both spec limits.
Cpk	Process capability index — centring-aware. Cpk = min((USL − μ) / (3σ_w), (μ − LSL) / (3σ_w)). Uses within-subgroup sigma.
Cpm	Taguchi capability index. Cpm = (USL − LSL) / (6√(s² + (μ − T)²)) where T is the target (spec midpoint).
Pp	Process performance ratio — spread-only index using overall SD. Pp = (USL − LSL) / (6s). Requires both spec limits.
Ppk	Process performance index — centring-aware. Ppk = min((USL − μ) / (3s), (μ − LSL) / (3s)). Uses overall SD.
Percentile method	Non-parametric capability estimation per ISO 22514-2: P0.135 and P99.865 sample quantiles are used as the ±3σ equivalents for non-normal data.
sigma_w	Within-subgroup sigma estimate: MR-bar / d2, where MR-bar is the average moving range of consecutive individual observations and d2 = 1.128 (n=2 subgroups).
Moving range	The absolute difference between consecutive observations: MR_i = |x_i − x_{i-1}|. Used to estimate short-term process variation.
PPM	Parts per million non-conforming — estimated from the normal distribution tails beyond the spec limits using within-subgroup sigma.
Sigma level	Cpk × 3 — the number of within-subgroup sigma units separating the process mean from the nearest spec limit.
Shapiro-Wilk	W test for normality; H0 = normal. p ≥ 0.05 indicates approximate normality; p < 0.05 indicates

	departure from normality.
OOC	Out of control — a data point beyond the 3σ control limits on an I or MR chart.
IQ	Installation Qualification
OQ	Operational Qualification
PQ	Performance Qualification
UR	User Requirement
TC	Test Case
jrrun	The validated bash entry-point script that sets the controlled environment before executing any script
CSV	Comma-separated values input file
PNG	Portable Network Graphics output file; plots saved to ~/Downloads/

5. User Requirements

The following user requirements define the expected behaviour of the Process Capability module. Each requirement is verifiable by the test cases in Section 7.

UR ID	Title	Requirement
UR-CAP-001	Input — CSV file	All Cap scripts shall accept a two-column CSV file with a header row. The file must be readable and contain at least one numeric column.
UR-CAP-002	Input — column name	All Cap scripts shall accept a column name argument that selects the measurement column from the CSV file.
UR-CAP-003	Input — LSL	All Cap scripts shall accept a numeric LSL or the literal '-' to indicate no lower specification limit.
UR-CAP-004	Input — USL	All Cap scripts shall accept a numeric USL or the literal '-' to indicate no upper specification limit.
UR-CAP-005	Input — at least one spec limit	All Cap scripts shall reject inputs where both LSL and USL are '-', with an informative error message.
UR-CAP-006	Input — LSL < USL	All Cap scripts shall reject inputs where the numeric LSL is greater than or equal to the numeric USL, with an informative error message.
UR-CAP-007	Input — minimum n	All Cap scripts shall require at least 5 valid observations and shall reject inputs with fewer, with an informative error message.
UR-CAP-008	Input — numeric column	All Cap scripts shall reject a specified column that contains no numeric values, with an informative error message.
UR-CAP-009	Input — file existence	All Cap scripts shall reject a non-existent input file with an informative error message.
UR-CAP-010	jrc_cap_normal — within-sigma Cpk	jrc_cap_normal shall compute Cpk using within-subgroup sigma estimated from the average moving range: σ_w = MR-bar / 1.128 (d2 for n=2 subgroups).
UR-CAP-011	jrc_cap_normal — overall-sigma Ppk	jrc_cap_normal shall compute Ppk using the overall sample standard deviation.
UR-CAP-012	jrc_cap_normal — Cp and Pp	jrc_cap_normal shall compute Cp and Pp when both LSL and USL are provided. These indices shall not be reported for one-sided analyses.
UR-CAP-013	jrc_cap_normal — Cpm	jrc_cap_normal shall compute Cpm (Taguchi index) when both spec limits are provided, using the spec midpoint as the target value.
UR-CAP-014	jrc_cap_normal — capability verdict	jrc_cap_normal shall report a categorical verdict based on Cpk: EXCELLENT (≥ 1.67), CAPABLE (≥ 1.33), MARGINAL (≥ 1.00), NOT CAPABLE (< 1.00).
UR-CAP-015	jrc_cap_nonnormal — percentile Ppk	jrc_cap_nonnormal shall compute Pp and Ppk using the percentile method (ISO 22514-2): P0.135 and P99.865 sample quantiles used as ±3σ equivalents.
UR-CAP-016	jrc_cap_nonnormal — normality advisory	jrc_cap_nonnormal shall perform a Shapiro-Wilk normality test and display an advisory suggesting jrc_cap_normal when p ≥ 0.05.
UR-CAP-017	jrc_cap_nonnormal — observed non-conformance	jrc_cap_nonnormal shall report the observed percentage of measurements below LSL and above USL.
UR-CAP-018	jrc_cap_sixpack — six-	jrc_cap_sixpack shall produce a single PNG containing six panels:

	panel PNG	Individuals chart, Moving Range chart, capability histogram, normal probability plot, numerical summary, and colour-coded verdict.
UR-CAP-019	jrc_cap_sixpack — OOC detection	jrc_cap_sixpack shall flag observations beyond the 3σ I-chart control limits (UCL_X, LCL_X) and moving ranges beyond UCL_MR as OOC signals.
UR-CAP-020	Output — PNG to ~/Downloads/	All Cap scripts shall save a PNG output to ~/Downloads/ with a YYYYMMDD_HHMMSS timestamp prefix and script-specific filename suffix.
UR-CAP-021	Bypass protection — RENV_PATHS_ROOT	All Cap scripts shall check the RENV_PATHS_ROOT environment variable at startup and call stop() with an informative message if it is not set, preventing execution outside the validated jrrun environment.
UR-CAP-022	Error handling — non-zero exit	All Cap scripts shall exit with a non-zero status code whenever an error condition is detected, and shall print an informative message to stderr.

6. Test Environment

Component	Specification
Operating System	macOS 15.x (Apple Silicon) or Windows 10/11 (Git Bash)
R Version	As installed per JR-IQ-001 (R 4.3.x or later)
Python Version	As installed per JR-IQ-001 (Python 3.11.x or later)
Test Framework	pytest 8.x
Test runner	repos/cap/admin_cap_oq
Test data	repos/cap/oq/data/ (5 CSV fixtures)
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (shared with other module OQ suites)
Output directory	~/Downloads/ (PNG output verified by mtime glob)
Entry point	bin/jrrun (integrity-checked execution)

7. Test Cases

All test cases are implemented in repos/cap/oq/test_cap_normal.py, test_cap_nonnormal.py, and test_cap_sixpack.py. Test data files reside in repos/cap/oq/data/.

7.1 jrc_cap_normal

TC-CAP-N-001

Tests: UR-CAP-001, UR-CAP-010
Command: `jrrun jrc_cap_normal.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Valid input with both spec limits.
Pass criterion: exit 0; 'Capability' in output

TC-CAP-N-002

Tests: UR-CAP-010
Command: `jrrun jrc_cap_normal.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Cpk label present.
Pass criterion: exit 0; 'Cpk:' in output

TC-CAP-N-003

Tests: UR-CAP-012
Command: `jrrun jrc_cap_normal.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Cp label present with both limits.
Pass criterion: exit 0; 'Cp:' in output

TC-CAP-N-004

Tests: UR-CAP-011

Command: `jrrun jrc_cap_normal.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Ppk label present.
Pass criterion: exit 0; 'Ppk:' in output

TC-CAP-N-005

Tests: UR-CAP-014
Command: `jrrun jrc_cap_normal.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Capable dataset — tight data well within spec.
Pass criterion: exit 0; 'CAPABLE' or 'EXCELLENT' in output

TC-CAP-N-006

Tests: UR-CAP-014
Command: `jrrun jrc_cap_normal.R cap_normal_marginal.csv value 9.0 11.0`
Rationale: Shifted dataset — mean near USL.
Pass criterion: exit 0; 'MARGINAL' or 'NOT CAPABLE' in output

TC-CAP-N-007

Tests: UR-CAP-020
Command: `jrrun jrc_cap_normal.R cap_normal_capable.csv value 9.0 11.0`
Rationale: PNG output to ~/Downloads/.
Pass criterion: exit 0; *_jrc_cap_normal.png found in ~/Downloads/

TC-CAP-N-008

Tests: UR-CAP-022
Command: `jrrun jrc_cap_normal.R  (no arguments)`
Rationale: Missing arguments.
Pass criterion: exit non-zero; 'Usage' in output

TC-CAP-N-009

Tests: UR-CAP-009
Command: `jrrun jrc_cap_normal.R nonexistent.csv value 9.0 11.0`
Rationale: Non-existent file.
Pass criterion: exit non-zero

TC-CAP-N-010

Tests: UR-CAP-007
Command: `jrrun jrc_cap_normal.R cap_small.csv value 9.0 11.0`
Rationale: n = 3 (< 5).
Pass criterion: exit non-zero

TC-CAP-N-011

Tests: UR-CAP-008
Command: `jrrun jrc_cap_normal.R cap_nonnumeric.csv value 0.0 1.0`
Rationale: Non-numeric column.
Pass criterion: exit non-zero

TC-CAP-N-012

Tests: UR-CAP-006
Command: `jrrun jrc_cap_normal.R cap_normal_capable.csv value 11.0 9.0`
Rationale: LSL > USL.
Pass criterion: exit non-zero

TC-CAP-N-013

Tests: UR-CAP-021
Command: `Rscript --vanilla jrc_cap_normal.R ... (RENV_PATHS_ROOT unset)`

Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

7.2 jrc_cap_nonnormal

TC-CAP-NN-001

Tests: UR-CAP-001, UR-CAP-015
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 0.0 6.0`
Rationale: Valid non-normal input with both spec limits.
Pass criterion: exit 0; 'Capability' in output

TC-CAP-NN-002

Tests: UR-CAP-015
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 0.0 6.0`
Rationale: Ppk label present.
Pass criterion: exit 0; 'Ppk' in output

TC-CAP-NN-003

Tests: UR-CAP-015
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 0.0 6.0`
Rationale: P0.135 label present.
Pass criterion: exit 0; 'P0.135' in output

TC-CAP-NN-004

Tests: UR-CAP-015
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 0.0 6.0`
Rationale: P99.865 label present.
Pass criterion: exit 0; 'P99.865' in output

TC-CAP-NN-005

Tests: UR-CAP-016
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 0.0 6.0`
Rationale: Shapiro-Wilk result reported.
Pass criterion: exit 0; 'Shapiro-Wilk' in output

TC-CAP-NN-006

Tests: UR-CAP-016
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 0.0 6.0`
Rationale: Skewed data should give SW $p < 0.05$ (no normality advisory).
Pass criterion: exit 0; 'Data may be approximately normal' NOT in output

TC-CAP-NN-007

Tests: UR-CAP-020
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 0.0 6.0`
Rationale: PNG output to ~/Downloads/.
Pass criterion: exit 0; *_jrc_cap_nonnormal.png found in ~/Downloads/

TC-CAP-NN-008

Tests: UR-CAP-022
Command: `jrrun jrc_cap_nonnormal.R  (no arguments)`
Rationale: Missing arguments.
Pass criterion: exit non-zero; 'Usage' in output

TC-CAP-NN-009

Tests: UR-CAP-009
Command: `jrrun jrc_cap_nonnormal.R nonexistent.csv value 0.0 6.0`
Rationale: Non-existent file.
Pass criterion: exit non-zero

TC-CAP-NN-010

Tests: UR-CAP-007
Command: `jrrun jrc_cap_nonnormal.R cap_small.csv value 0.0 20.0`
Rationale: n = 3 (< 5).
Pass criterion: exit non-zero

TC-CAP-NN-011

Tests: UR-CAP-008
Command: `jrrun jrc_cap_nonnormal.R cap_nonnumeric.csv value 0.0 1.0`
Rationale: Non-numeric column.
Pass criterion: exit non-zero

TC-CAP-NN-012

Tests: UR-CAP-006
Command: `jrrun jrc_cap_nonnormal.R cap_nonnormal.csv value 6.0 0.0`
Rationale: LSL > USL.
Pass criterion: exit non-zero

TC-CAP-NN-013

Tests: UR-CAP-021
Command: `Rscript --vanilla jrc_cap_nonnormal.R ... (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

7.3 jrc_cap_sixpack

TC-CAP-S-001

Tests: UR-CAP-001, UR-CAP-018
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Valid input with both spec limits.
Pass criterion: exit 0; 'Sixpack' in output

TC-CAP-S-002

Tests: UR-CAP-010
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Cpk label present.
Pass criterion: exit 0; 'Cpk:' in output

TC-CAP-S-003

Tests: UR-CAP-011
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Ppk label present.
Pass criterion: exit 0; 'Ppk:' in output

TC-CAP-S-004

Tests: UR-CAP-019
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: UCL and LCL reported for I-MR chart.
Pass criterion: exit 0; 'UCL' and 'LCL' in output

TC-CAP-S-005

Tests: UR-CAP-016
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Shapiro-Wilk result reported.
Pass criterion: exit 0; 'Shapiro-Wilk' in output

TC-CAP-S-006

Tests: UR-CAP-018
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: SPC verdict present in output.
Pass criterion: exit 0; 'SPC:' in output

TC-CAP-S-007

Tests: UR-CAP-014, UR-CAP-018
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: Capability verdict present.
Pass criterion: exit 0; 'Cap:' in output; one of CAPABLE/EXCELLENT/MARGINAL/NOT CAPABLE in output

TC-CAP-S-008

Tests: UR-CAP-020
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 9.0 11.0`
Rationale: PNG output to ~/Downloads/.
Pass criterion: exit 0; *_jrc_cap_sixpack.png found in ~/Downloads/

TC-CAP-S-009

Tests: UR-CAP-022
Command: `jrrun jrc_cap_sixpack.R  (no arguments)`
Rationale: Missing arguments.
Pass criterion: exit non-zero; 'Usage' in output

TC-CAP-S-010

Tests: UR-CAP-009
Command: `jrrun jrc_cap_sixpack.R nonexistent.csv value 9.0 11.0`
Rationale: Non-existent file.
Pass criterion: exit non-zero

TC-CAP-S-011

Tests: UR-CAP-007
Command: `jrrun jrc_cap_sixpack.R cap_small.csv value 9.0 11.0`
Rationale: n = 3 (< 5).
Pass criterion: exit non-zero

TC-CAP-S-012

Tests: UR-CAP-008
Command: `jrrun jrc_cap_sixpack.R cap_nonnumeric.csv value 0.0 1.0`
Rationale: Non-numeric column.
Pass criterion: exit non-zero

TC-CAP-S-013

Tests: UR-CAP-006
Command: `jrrun jrc_cap_sixpack.R cap_normal_capable.csv value 11.0 9.0`
Rationale: LSL > USL.
Pass criterion: exit non-zero

TC-CAP-S-014

Tests: UR-CAP-021
Command: `Rscript --vanilla jrc_cap_sixpack.R ... (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

8. Requirements Traceability Matrix

Each user requirement is covered by at least one test case.

UR ID	Title	Test Cases
UR-CAP-001	All Cap scripts accept CSV input	TC-CAP-N-001, TC-CAP-NN-001, TC-CAP-S-001
UR-CAP-002	Column name argument	TC-CAP-N-001..007, TC-CAP-NN-001..007, TC-CAP-S-001..008 (all run with col='value')
UR-CAP-003	LSL argument	TC-CAP-N-001..007, TC-CAP-NN-001..007, TC-CAP-S-001..008
UR-CAP-004	USL argument	TC-CAP-N-001..007, TC-CAP-NN-001..007, TC-CAP-S-001..008
UR-CAP-005	Both limits omitted rejected	Covered by input guard in all scripts (LSL='-', USL='-' tested in pytest args)
UR-CAP-006	LSL >= USL rejected	TC-CAP-N-012, TC-CAP-NN-012, TC-CAP-S-013
UR-CAP-007	n < 5 rejected	TC-CAP-N-010, TC-CAP-NN-010, TC-CAP-S-011
UR-CAP-008	Non-numeric column rejected	TC-CAP-N-011, TC-CAP-NN-011, TC-CAP-S-012
UR-CAP-009	Non-existent file rejected	TC-CAP-N-009, TC-CAP-NN-009, TC-CAP-S-010
UR-CAP-010	Normal — Cpk via within sigma	TC-CAP-N-002, TC-CAP-S-002
UR-CAP-011	Normal — Ppk via overall sigma	TC-CAP-N-004, TC-CAP-S-003
UR-CAP-012	Normal — Cp and Pp with both limits	TC-CAP-N-003
UR-CAP-013	Normal — Cpm with both limits	TC-CAP-N-001 (Cpm: present in output)
UR-CAP-014	Normal — capability verdict	TC-CAP-N-005, TC-CAP-N-006, TC-CAP-S-007
UR-CAP-015	Non-normal — percentile Ppk	TC-CAP-NN-002, TC-CAP-NN-003, TC-CAP-NN-004
UR-CAP-016	Non-normal — normality advisory	TC-CAP-NN-005, TC-CAP-NN-006, TC-CAP-S-005
UR-CAP-017	Non-normal — observed non-conformance	TC-CAP-NN-001 (output verified for both limit cases)
UR-CAP-018	Sixpack — six-panel PNG	TC-CAP-S-001, TC-CAP-S-006, TC-CAP-S-007
UR-CAP-019	Sixpack — OOC detection	TC-CAP-S-004 (UCL/LCL present in output)
UR-CAP-020	PNG to ~/Downloads/	TC-CAP-N-007, TC-CAP-NN-007, TC-CAP-S-008
UR-CAP-021	Bypass protection	TC-CAP-N-013, TC-CAP-NN-013, TC-CAP-S-014
UR-CAP-022	Non-zero exit on error	TC-CAP-N-008..012, TC-CAP-NN-008..012, TC-CAP-S-009..013

9. Acceptance Criteria

The Process Capability module OQ is considered PASS when all of the following conditions are satisfied:

- All 40 automated test cases execute with the result PASSED.
- Zero test cases result in FAILED or ERROR.
- The OQ evidence file is generated and preserved at ~/.jrscript/<PROJECT_ID>/validation/cap_oq_execution_<timestamp>.txt.
- All test cases are executed using the validated jrrun entry-point (bypass protection verified).
- No open deviations remain unresolved at the time of approval.

Any test failure shall be investigated, documented as a deviation, resolved, and the affected test(s) re-executed. The OQ may not be closed until all deviations are resolved.